

Business & Use-Case Specification

ServiceGuard AI

UC-01: Identify services at risk

```
Engineer → Dashboard
↓
Request at-risk services
↓
Backend retrieves current service health
↓
ML/rules rank risk
↓
Dashboard shows prioritized services
```

UC-02: Investigate an incident

```
Incident
↓
Telemetry evidence
↓
Dependency/environment evidence
↓
ML risk
↓
Runbook retrieval
↓
AI explanation
```

UC-03: Ask operational question

Example: "Why is Payment API at risk and what should I do first?" The agent determines which tools are needed, gathers evidence, and returns a structured answer with evidence.

UC-04: Approve recommendation

- Engineer reviews recommendation.
- Engineer sees supporting telemetry and runbook evidence.
- Engineer approves/rejects/modifies.
- Decision and actor are recorded in audit history.

UC-05: Service degradation scenario

```
Latency 180ms → 310ms → 820ms → 1800ms
Error rate 0.5% → 1.2% → 4.8% → 8.4%
CPU 42% → 63% → 81% → 91%
DB connections 48 → 52 → 71 → 89 → 95%
↓
Incident detected
↓
ML risk = HIGH
↓
Runbook retrieved
↓
AI recommends DB investigation
↓
Engineer approves
```

Business value

- Prioritized incident response.
- Reduced investigation time.

- Consistent use of operational procedures.
- Better explainability than a chatbot-only system.
- Auditable AI-assisted operations.